YUNYANG LIU

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SUMMARY

New Grad Robotics Engineering candidate specializing in mechatronic system design, embedded control, CAD, and rapid prototyping. Proven ability to take robotic systems from concept to working hardware through iterative design, fabrication, and testing.

EDUCATION

University of California, Santa Cruz
Robotics Engineering & Electrical Engineering
Expected Jun 2026 · GPA: 3.58 / 4.0

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB
Embedded Systems: PIC32MX, STM32F411RE (Nucleo), ESP32-C3, BASYS3 FPGA
Hardware & Interfaces: UART, SPI, I²C, PWM, ADC/DAC, GPIO, Interrupts
Tools: Onshape, MPLAB X, PlatformIO, VS Code, Vivado, MATLAB, Git/GitHub, Linux CLI
Fabrication: FDM 3D Printing (Bambu Lab P1S), Laser Cutting, PCB Soldering

EXPERIENCE

Assistive 7-DoF Robotic Manipulator for Remote Operation *Sep 2025 – Jun 2026*

- Led full mechanical design (Onshape), FDM prototyping (Bambu Lab P1S), and assembly of a 7-DoF assistive robotic arm. Iterated 5+ times to refine joint tolerances and structural geometry, achieving a reliable full-arm assembly with 1.33 kg payload capacity.
- Calibrated servo performance (neutral alignment, response consistency) to improve positioning accuracy and ensure coordinated motion across all joints.
- Served as Mechanical Lead in a 4-person cross-functional team, aligning mechanical design decisions with electrical and software subsystems.
- Developed and executed system-level tests to evaluate motion accuracy, load capacity, and repeatability; identified tolerance stack-up issues and drove iterative design improvements.

4-DoF Sensor-Driven Robotic Manipulator (IMU + Flex Sensors) *Jan – Mar 2026*

- Architected a gesture-controlled 4-DoF robotic arm using STM32F411RE, integrating IMU, flex, and FSR sensors for PWM-driven actuation; achieved reliable pick-and-place in <30s (2× faster than requirement) with robust real-time sensor-to-actuation contro
- Designed and fabricated mechanical components via parametric CAD (Onshape) and FDM 3D printing; iterated on designs to resolve assembly, servo clearance, and alignment issues, improving structural reliability
- Developed a power distribution system with dual buck converters and per-servo decoupling, eliminating voltage drops under >6A stall current for stable multi-servo operation
- Executed system-level tests to evaluate gesture tracking accuracy, response latency, and control stability; diagnosed sensor noise and calibration issues, and improved system robustness through iterative tuning

Sensor-Driven Embedded System Design *Jan – Mar 2026*

- Developed embedded firmware in C on STM32F411RE, interfacing multi-sensor systems (IMU, encoder, ultrasonic, flex) via ADC, I2C, PWM, and interrupt-driven architecture to enable real-time sensor-driven control.
- Designed signal processing pipelines (IIR filtering, regression) to suppress sensor noise, improving stability of sensor-to-actuator mapping under dynamic conditions.
- Built interrupt-driven state machines for encoder tracking and ultrasonic sensing, boosting measurement accuracy and achieving 3.75° rotational resolution.
- Calibrated IMU sensors using bias/scale correction and ellipsoid fitting, reducing drift and significantly improving orientation accuracy and system stability.

Autonomous Event-Driven Card Dispensing System (PWM Motor Control & Sensor Feedback) *Apr – Jun 2025*

- Architected a hierarchical state machine (HSM) in C (ES Framework) to coordinate servo scanning, player detection, and multi-player card dispensing, enabling robust real-time control.
- Integrated ultrasonic sensing, servo positioning, and DC motor control (L298 H-bridge) into a cohesive embedded system on a custom-soldered perfboard, ensuring reliable cross-component synchronization.
- Diagnosed and resolved multi-card dispensing failures via mechanical and control optimizations (friction guides, reverse motor logic), achieving consistent single-card output and improving system reliability.
- Designed and fabricated a custom chassis using parametric CAD (Onshape), combining FDM 3D printing and laser cutting with kerf compensation to ensure precise assembly and repeatable alignment.

Two-Wheeled Autonomous Mobile Robot *Apr – Jun 2025*

- Built a keyboard-driven control interface for manual override and debugging, enabling real-time command input and stable, responsive robot control.

- Developed reactive autonomy using HSM/FSM in C (ES Framework) on PIC32, integrating bump sensors and IR phototransistors to achieve light-seeking behavior, collision-triggered direction correction, and low-light automatic stop.

Full-Duplex UART Communication Stack with Packet Protocol *Jan – Mar 2025*

- Implemented an interrupt-driven UART driver on PIC32 with dual circular FIFOs and a custom packet protocol (header/length/ID/payload/tail + checksum), enabling reliable full-duplex communication with validated data integrity.
- Designed an RX state machine for packet parsing and checksum verification, filtering invalid data and improving communication robustness.
- Interfaced an AS5047D magnetic encoder via SPI, decoding 14-bit angle data using a two-transaction protocol and configuring clock phase/polarity, achieving stable and accurate angle readings.

Real Estate Development Intern | Presidio Bay Ventures, San Francisco *Jun – Aug 2023*

- Conducted 10+ cross-functional stakeholder interviews to assess feasibility, regulatory constraints, and financial considerations for mixed-use development projects
- Developed and presented a 45-slide mixed-use development proposal by synthesizing site analysis and stakeholder insights, supporting executive-level decision-making